

Roll No LHR-91-11-18 (To be filled in by the candidate)**MATHEMATICS** (Academic Sessions 2014 – 2016 to 2017 – 2019)

Q.PAPER – I (Objective Type) 218-(INTER PART – I)

Time Allowed : 30 Minutes

GROUP – I

Maximum Marks : 20

PAPER CODE = 6191

Note : Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer-book. Cutting or filling two or more circles will result in zero mark in that question.

1-1	The set $\{0, 1\}$ is closed under : (A) Addition (B) Multiplication (C) Division (D) Subtraction
2	If A and B are two sets, then $A - B =$: (A) $A \cup B^c$ (B) $(A \cup B)^c$ (C) $A \cap B^c$ (D) $(A \cap B)^c$
3	A square matrix A is skew symmetric if $A^t =$: (A) $-A$ (B) A (C) $\overline{A}$ (D) A^t
4	If order of a matrix A is $m \times n$, then order of A^t is : (A) $m \times n$ (B) $m \times m$ (C) $n \times m$ (D) $n \times n$
5	Sum of roots of quadratic equation $ax^2 + bx + c = 0$ is : (A) $\frac{a}{b}$ (B) $\frac{b}{a}$ (C) $\frac{c}{a}$ (D) $-\frac{b}{a}$
6	Product of all fourth roots of unity is : (A) -1 (B) 0 (C) 1 (D) i
7	The fraction $\frac{3x^2 + 5}{x + 1}$ is : (A) Proper fraction (B) Polynomial (C) Partial fraction (D) Improper fraction
8	Geometric mean between -2 and 8 is : (A) 4 (B) ± 4 (C) 8 (D) $\pm 4i$
9	The 10th term of $\frac{1}{2}, \frac{1}{5}, \frac{1}{8}, \dots$ is : (A) 30 (B) 28 (C) $\frac{1}{29}$ (D) $\frac{1}{32}$
10	The value of $\frac{4!}{0!}$ is : (A) 24 (B) 4 (C) 0 (D) Infinity
11	If A and B are mutually exclusive events, then $P(A \cup B) =$: (A) $P(A) \cup P(B)$ (B) $P(A) + P(B)$ (C) $P(A \cap B)$ (D) $P(A) - P(B)$

(Turn Over)

LHR-G1-11-18

(2)

1-12	$4^n > 3^n + 4$ is true for integral values of $n =$: (A) 1 (B) $n \leq 1$ (C) 0 (D) $n \geq 2$
13	The 2 nd term in expansion of $\left(1 - \frac{1}{3}x\right)^{-1}$ is : (A) $\frac{1}{3}x$ (B) $-\frac{1}{3}x$ (C) $3x$ (D) $2x$
14	If $\sin \theta < 0$ and $\cot \theta > 0$, then θ lies in quadrant : (A) 1 (B) 2 (C) 3 (D) 4
15	If α, β, γ are angles of triangle then $\tan(\alpha + \beta) + \tan \gamma =$: (A) 1 (B) 0 (C) 2 (D) -1
16	Period of $\cos\left(\frac{x}{2}\right) =$: (A) 2π (B) $\frac{\pi}{2}$ (C) 3π (D) 4π
17	Radius of escribed circle opposite to vertex 'c' of the triangle is : (A) $\frac{\Delta}{s}$ (B) $\frac{\Delta}{s-a}$ (C) $\frac{\Delta}{s-c}$ (D) $\frac{\Delta}{s-b}$
18	The value escribed circle $r_1 =$: (A) $\frac{\Delta}{s-a}$ (B) $\frac{\Delta}{s-c}$ (C) $\frac{\Delta}{s}$ (D) $\frac{\Delta}{a}$
19	The value of $\cos(\tan^{-1} 0) =$: (A) -1 (B) 1 (C) 0 (D) ∞
20	If $\cos x = -\frac{1}{2}$, then reference angle is : (A) $\frac{\pi}{6}$ (B) $-\frac{\pi}{3}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{2}$

Roll No LHR-41-11-18 (To be filled in by the candidate)

(Academic Sessions 2014 – 2016 to 2017 – 2019) .

MATHEMATICS

218-(INTER PART – I)

Time Allowed : 2.30 hours

PAPER – I (Essay Type)

GROUP – I

Maximum Marks : 80

SECTION – I

2. Write short answers to any EIGHT (8) questions :

16

- (i) Simplify $(-1)^{-21}$
- (ii) Express the complex number $(1 + i\sqrt{3})$ in polar form.
- (iii) Find the multiplicative inverse of $(-4, 7)$
- (iv) Is there any set which has no proper subset? If so name that set.
- (v) Write the converse and contrapositive of $\sim q \rightarrow \sim p$
- (vi) For $A = \{1, 2, 3, 4\}$, find the relation in A for $R = \{(x, y) | x + y < 5\}$, also write the range of R.
- (vii) If $A = \begin{bmatrix} 1 & 2 \\ a & b \end{bmatrix}$, $A^2 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$, find the values of a and b .
- (viii) Find the multiplicative inverse of the matrix $\begin{bmatrix} 2i & i \\ i & -i \end{bmatrix}$
- (ix) Show that $\begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ yz & zx & xy \end{vmatrix} = \begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ x^2 & y^2 & z^2 \end{vmatrix}$
- (x) Solve the equation $x^4 - 6x^2 + 8 = 0$
- (xi) Show that $x^3 - y^3 = (x - y)(x - \omega y)(x - \omega^2 y)$, ω is complex cube root of unity.
- (xii) If α, β are the roots of $3x^2 - 2x + 4 = 0$, then find the value of $\frac{1}{\alpha^3} + \frac{1}{\beta^3}$

3. Write short answers to any EIGHT (8) questions :

16

- (i) Resolve $\frac{x^2 + 1}{(x + 1)(x - 1)}$ into partial fractions.
- (ii) If $a_{n-2} = 3n - 11$, find the n th term of the sequence
- (iii) If 5, 8 are two A.Ms between a and b , find a and b
- (iv) Which term of the A.P. 5, 2, -1, ----- is - 85 ?
- (v) Insert two G.Ms between 1 and 8.
- (vi) If 5 is the harmonic mean between 2 and b , find b
- (vii) Define fundamental principle of counting.
- (viii) Find the number of the diagonals of a 6-sided figure.
- (ix) What is probability that a slip of numbers divisible by 4 are picked from the slips bearing number 1, 2, 3, ----- 10?
- (x) State the principle of mathematical induction.
- (xi) If x is so small that its square and higher powers can be neglected, then show that $\frac{1-x}{\sqrt{1+x}} = 1 - \frac{3}{2}x$
- (xii) Find the 6th term in the expansion of $\left(x^2 - \frac{3}{2x}\right)^{10}$

4. Write short answers to any NINE (9) questions :

18

- (i) An arc subtends an angle of 70° at the center of a circle and its length is 132 m. Find the radius of the circle.
- (ii) Define coterminal angles.
- (iii) Verify $\sin^2 \frac{\pi}{6} + \sin^2 \frac{\pi}{3} + \tan^2 \frac{\pi}{4} = 2$
- (iv) If α, β, γ are angles of a triangle ΔABC , then prove that $\tan(\alpha + \beta) + \tan \gamma = 0$
- (v) Find the value of $\sin 105^\circ$, without calculator.
- (vi) Prove that $\cot \alpha - \tan \alpha = 2 \cot 2\alpha$
- (vii) Write the domain of $y = \sin x$
- (viii) A vertical pole is 8m high and the length of its shadow is 6m. What is the angle of elevation of the sun at that moment?
- (ix) Find α and β in the triangle ΔABC in which $a = 7$, $b = 7$, $c = 9$
- (x) Find the area of the triangle ΔABC in which $a = 200$, $b = 120$, $\gamma = 150^\circ$
- (xi) Evaluate without using calculator $\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$
- (xii) Solve the equation $2\sin x - 1 = 0$
- (xiii) Find the solution of the equation which lie in interval $[0, 2\pi]$: $\sec x = -2$

SECTION - II

Note : Attempt any THREE questions.

5. (a) Consider the set $S = \{1, -1, i, -i\}$. Set up its multiplication table and show that the set S is an abelian group under multiplication.

5

- (b) If $A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 1 & -1 & 1 \end{bmatrix}$ then find A^{-1} by using adjoint of the matrix.

5

6. (a) Solve the system of equations : $x + y = a + b$; and $\frac{a}{x} + \frac{b}{y} = 2$

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- (b) Resolve $\frac{9x-7}{(x^2+1)(x+3)}$ into partial fractions.

5

7. (a) Find four numbers in arithmetic sequence (A.P.) whose sum is 32 and the sum of whose squares is 276.

5

- (b) Use binomial series to show that $1 + \frac{1}{4} + \frac{1 \times 3}{4 \times 8} + \frac{1 \times 3 \times 5}{4 \times 8 \times 12} + \dots = \sqrt{2}$

5

8. (a) If $\operatorname{cosec} \theta = \frac{m^2+1}{2m}$ and $m > 0$ $\left(0 < \theta < \frac{\pi}{2}\right)$, find the values of the all remaining trigonometric ratios.

5

- (b) Prove that $\sin \frac{\pi}{9} \sin \frac{2\pi}{9} \sin \frac{\pi}{3} \sin \frac{4\pi}{9} = \frac{3}{16}$ without using calculator.

5

9. (a) With usual notations, prove that $r_1 = \frac{\Delta}{s}$

5

- (b) Prove that $\sin^{-1} \frac{3}{5} + \sin^{-1} \frac{8}{17} = \sin^{-1} \frac{77}{85}$

5

Roll No CHR-G2-1118 (To be filled in by the candidate)

MATHEMATICS (Academic Sessions 2014 – 2016 to 2017 – 2019)

Q.PAPER – I (Objective Type) 218-(INTER PART – I)

Time Allowed : 30 Minutes

GROUP – II

Maximum Marks : 20

PAPER CODE = 6194

Note : Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer-book. Cutting or filling two or more circles will result in zero mark in that question.

1-1	$2 \sin\left(\frac{P+Q}{2}\right) \cos\left(\frac{P-Q}{2}\right) = \text{-----} :$ (A) $\sin P + \sin Q$ (B) $\sin P - \sin Q$ (C) $\cos P + \cos Q$ (D) $\cos P - \cos Q$
2	With usual notation ${}^nC_0 = :$ (A) 1 (B) 0 (C) n (D) 2
3	$\sin^{-1} A - \sin^{-1} B = \text{-----} :$ (A) $\sin^{-1}(A\sqrt{1-B^2} - B\sqrt{1-A^2})$ (B) $\sin^{-1}(A\sqrt{1-B^2} + B\sqrt{1-A^2})$ (C) $\cos^{-1}(A\sqrt{1-B^2} - B\sqrt{1-A^2})$ (D) $\cos^{-1}(A\sqrt{1-B^2} + B\sqrt{1-A^2})$
4	Values of trigonometric functions of the quadrantal angle 765° are same as of the angle : (A) 30° (B) 45° (C) 60° (D) 90°
5	Solution of $\cot \theta = \frac{1}{\sqrt{3}}$ in quadrant – III : (A) $\frac{5\pi}{4}$ (B) $\frac{7\pi}{6}$ (C) $\frac{4\pi}{3}$ (D) π
6	The sum of coefficients in the binomial expansion when $n = 4$ is : (A) 1 (B) 8 (C) 16 (D) 32
7	With usual notation the “circum-radius” $R = \text{-----} :$ (A) $\frac{\Delta}{s}$ (B) $\frac{abc}{4\Delta}$ (C) $\frac{\Delta}{abc}$ (D) $\frac{s}{\Delta}$
8	Period of $3\sin 2x$ is : (A) 6π (B) 2π (C) π (D) $\frac{\pi}{2}$
9	Which one is divisible by 2 for all positive integral values of n : (A) $n^3 - n$ (B) $5^n - 1$ (C) $5^n - 2^n$ (D) $n^2 + n$
10	In law of tangents $\frac{\tan\left(\frac{\beta-\gamma}{2}\right)}{\tan\left(\frac{\beta+\gamma}{2}\right)} = :$ (A) $\frac{a-b}{a+b}$ (B) $\frac{c-a}{c+a}$ (C) $\frac{c-b}{c+b}$ (D) $\frac{b-c}{b+c}$
11	If ' ω ' be the cube root of unity, then $\omega^2 = :$ (A) $\frac{-1-\sqrt{3}i}{2}$ (B) $\frac{1-\sqrt{3}i}{2}$ (C) 1 (D) $\frac{1+\sqrt{3}i}{2}$

(Turn Over)

LHR-62-11-18 (2)

1-12	Multiplicative inverse of complex number $-3 - 5i$ is : (A) $\frac{3}{34} + \frac{5}{34}i$ (B) $\frac{-3}{34} - \frac{5}{34}i$ (C) $\frac{-3}{34} + \frac{5}{34}i$ (D) $\frac{-3}{\sqrt{34}} + \frac{5}{\sqrt{34}}i$
13	Simplify form of $\frac{10!}{7!}$ is equal to : (A) 720 (B) 620 (C) 520 (D) 420
14	If matrix $\begin{bmatrix} x & 4 \\ 2 & 8 \end{bmatrix}$ is singular then $x =$: (A) 0 (B) -1 (C) 2 (D) 1
15	Geometric mean between 4 and 16 are : (A) 10 (B) ± 8 (C) $\frac{32}{5}$ (D) 64
16	Roots of the equation $x^2 - 7x + 10 = 0$ are : (A) (2, -5) (B) (-2, 5) (C) (2, 5) (D) (-2, -5)
17	Formula for the sum of n terms of A.P. (Arithmetic progression) : (A) $a_n = a_1 + (n-1)d$ (B) $s_n = \frac{n}{2}(a_1 + a_n)$ (C) $s_n = \frac{a_1(1-r^n)}{1-r}$ (D) $s = \frac{a}{1-r}$
18	Tabular form of $\{x x \in E, 4 < x\}$: (A) $\{\}$ (B) $\{4\}$ (C) $\{6\}$ (D) $\{4, 6\}$
19	Partial fractions of $\frac{1}{(x^2 + 1)(x - 1)}$ are of the form : (A) $\frac{A}{x^2 + 1} + \frac{B}{x - 1}$ (B) $\frac{A}{x + 1} + \frac{B}{(x^2 + 1)} + \frac{C}{x - 1}$ (C) $\frac{A}{x^2 + 1} + \frac{Bx + C}{x - 1}$ (D) $\frac{Ax + B}{x^2 + 1} + \frac{C}{x - 1}$
20	A matrix A is said to be symmetric if : (A) $A^t = -A$ (B) $A^t = A$ (C) $(\bar{A})^t = A$ (D) $(\bar{A})^t = -A$

Roll No. LHR-G2-11-18 (To be filled in by the candidate)

(Academic Sessions 2014 – 2016 to 2017 – 2019)

MATHEMATICS 218-(INTER PART – I)

PAPER – I (Essay Type) GROUP – II

Time Allowed : 2.30 hours

Maximum Marks : 80

SECTION – I

2. Write short answers to any EIGHT (8) questions :

16

- (i) Does the set $\{1, -1\}$ close w.r.t. : (a) addition (b) multiplication
- (ii) Find multiplicative inverse of the complex number $(-4, 7)$
- (iii) If $z = 1 - i\sqrt{3}$, then find $|z|$
- (iv) Write inverse and contrapositive of $q \rightarrow p$
- (v) If $A = \{a, b, c\}$, then write all subsets of A and find $P(A)$
- (vi) Show that set of natural number is not a group w.r.t. addition.
- (vii) Define diagonal matrix with an example.
- (viii) If $A = \begin{bmatrix} 2 & 1 \\ 6 & 3 \end{bmatrix}$, then find A^{-1}
- (ix) Without expansion show that $\begin{vmatrix} 6 & 7 & 8 \\ 3 & 4 & 5 \\ 2 & 3 & 4 \end{vmatrix} = 0$
- (x) Find four 4th roots of unity.
- (xi) If α, β are roots of $x^2 - px - p - c = 0$, show that $(1 - \alpha)(1 + \beta) = c$
- (xii) Find quadratic equation whose roots are $2\omega, 2\omega^2$, where ω is cube roots of unity.

3. Write short answers to any EIGHT (8) questions :

16

- (i) Resolve $\frac{x^2 + 1}{(x + 1)(x - 1)}$ into partial fractions.
- (ii) Find the indicated term of the sequence 2, 6, 11, 17, ----- $a_7 = ?$
- (iii) Sum the series upto n-terms $\frac{1}{1 - \sqrt{x}} + \frac{1}{1 - x} + \frac{1}{1 + \sqrt{x}}$
- (iv) Insert two G.Ms between 1 and 8.
- (v) Find the sum of the infinite geometric series $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots$
- (vi) Find the 12th term of the harmonic sequence $\frac{1}{3}, \frac{2}{9}, \frac{1}{6}, \dots$
- (vii) Evaluate $\frac{15!}{15!(15-15)!}$
- (viii) Find the value of n, when $\frac{12 \times 11}{2!} = {}^nC_{10}$
- (ix) There are 5 green and 3 red balls in a box, one ball is taken out, find the probability that the ball drawn is green.
- (x) Find the number of the diagonals of a 6-sided figure.
- (xi) Find the term involving x^4 in the expansion of $(3 - 2x)^7$.
- (xii) Using binomial theorem find the value of $(1.03)^{\frac{1}{3}}$ upto three decimal places.

4. Write short answers to any NINE (9) questions :

18

- (i) Define angle in the standard position with figure.
- (ii) Find x, if $\tan^2 45^\circ - \cos^2 60^\circ = x \sin 45^\circ \cos 45^\circ \tan 60^\circ$
- (iii) Prove that $\frac{1}{1 + \sin \theta} - \frac{1}{1 - \sin \theta} = 2 \sec^2 \theta$

(Turn Over)

(2)

LHR-CL-11-18

4. (iv) Find the value of $\sin 540^\circ$ without using calculator.
- (v) Prove that $\tan\left(\frac{\pi}{4} - \theta\right) + \tan\left(\frac{3\pi}{4} + \theta\right) = 0$
- (vi) Express $\sin(x + 45^\circ)\sin(x - 45^\circ)$ as sum or difference.
- (vii) Find the period of $\cos \frac{x}{6}$
- (viii) Find the area of triangle ΔABC , in which $b = 37$, $c = 45$ and $\alpha = 30^\circ 50'$
- (ix) Prove that $r_1 r_2 r_3 = \Delta^2$ (Using usual notation)
- (x) Prove that $(r_1 + r_2) \tan \frac{\gamma}{2} = c$ (Using usual notation)
- (xi) Find domain and range of $y = \cos^{-1} x$
- (xii) Solve the equation $\sin x = \frac{1}{2}$
- (xiii) Find solutions of $\cot \theta = \frac{1}{\sqrt{3}}$ which lie in $[0, 2\pi]$

SECTION - II

Note : Attempt any THREE questions.

5. (a) Convert the following theorem to logical form and prove it by constructing truth table :
 $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

- (b) Solve the following system by reducing their augmented matrices to the echelon form :

$$x + 2y + z = 2$$

$$2x + y + 2z = -1$$

$$2x + 3y - z = 9$$

6. (a) If α, β are the roots of the equation $ax^2 + bx + c = 0$ then find the equation whose roots are $\frac{-1}{\alpha^3}, \frac{1}{\beta^3}$

- (b) Resolve $\frac{2x^4}{(x-3)(x+2)^2}$ into partial fraction.

7. (a) For what value of n , $\frac{a^n + b^n}{a^{n-1} + b^{n-1}}$ is the positive geometric mean (G.M.) between a and b

- (b) If x is so small that its square and higher powers can be neglected, then show that :

$$\frac{(1-x)^{\frac{1}{2}}(9-4x)^{\frac{1}{2}}}{(8+3x)^{\frac{1}{3}}} \approx \frac{3}{2} - \frac{61}{48}x.$$

8. (a) If $\operatorname{cosec} \theta = \frac{m^2 + 1}{2m}$ and $m > 0$, $\left(0 < \theta < \frac{\pi}{2}\right)$, find the values of the remaining trigonometric ratios.

- (b) Prove without using calculator that $\cos 20^\circ \cos 40^\circ \cos 60^\circ \cos 80^\circ = \frac{1}{16}$

9. (a) The sides of a triangle are $x^2 + x + 1$, $2x + 1$ and $x^2 - 1$. Prove that the greatest angle of the triangle is 120° .

- (b) Prove that $2 \tan^{-1} \frac{1}{3} + \tan^{-1} \frac{1}{7} = \frac{\pi}{4}$